Indian Mutual Fund Market Analysis

ETL Data Pipeline, Risk Metrics, and Performance Benchmarking

An End-to-End Analytics Framework and Interactive Dashboard Architecture

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Date: August 2026 | Version: 1.0

Executive Summary

The Indian Mutual Fund industry has grown rapidly, bringing higher market complexity, diverse product offerings, and large volumes of daily transactional data. Evaluating fund performance through simple absolute returns often masks underlying risk profiles, fee structures, and benchmark tracking errors.

This technical report presents an end-to-end mutual fund analytics framework. It combines a Python-based ETL data pipeline, quantitative risk analytics (including VaR and CVaR calculations), and an interactive dashboard architecture. By unifying 10 core datasets, spanning historical NAVs, AMC assets under management (AUM), expense ratios, and benchmark indices, this system automates data processing, risk calculation, and performance comparison across asset classes.

Data Sources and Ingestion Framework

Data Sources Inventory

The analytical platform ingests 10 core structured datasets to evaluate fund performance, cost efficiency, and risk-adjusted metrics:

File	Description
01_fund_master.csv	Primary schema mapping Fund IDs, Fund Names, Asset Classes (Equity, Debt, Hybrid), Sub-Categories, and Inception Dates.
02_nav_history.csv	Time-series dataset of daily Net Asset Values across Direct and Regular plans.
03_aum_by_fund_house.csv	Quarterly aggregate AUM tracked across Asset Management Companies (AMCs).
04_expense_ratios.csv	Historical Expense Ratios split by Direct vs. Regular schemes.
05_category_inflows.csv	Monthly net cash inflows and outflows aggregated by category.
06_fund_holdings.csv	Portfolio-level asset allocations, top equity/debt stock holdings, and concentration ratios.

07_risk_metrics.csv	Pre-calculated baseline metrics including Beta, Standard Deviation, and Sharpe Ratios.
08_dividend_history.csv	Historical payout distributions and dividend re-investment records.
09_cash_flows.csv	Inflows/outflows at the individual fund scheme level.
10_benchmark_indices.csv	Daily closing prices and total return indices (TRI) for benchmarks such as Nifty 50, Nifty Midcap 100, and Crisil Debt indices.

Ingestion Pipeline Architecture

The ingestion pipeline (data_ingestion.py) serves as the entry point for raw static and time-series files:

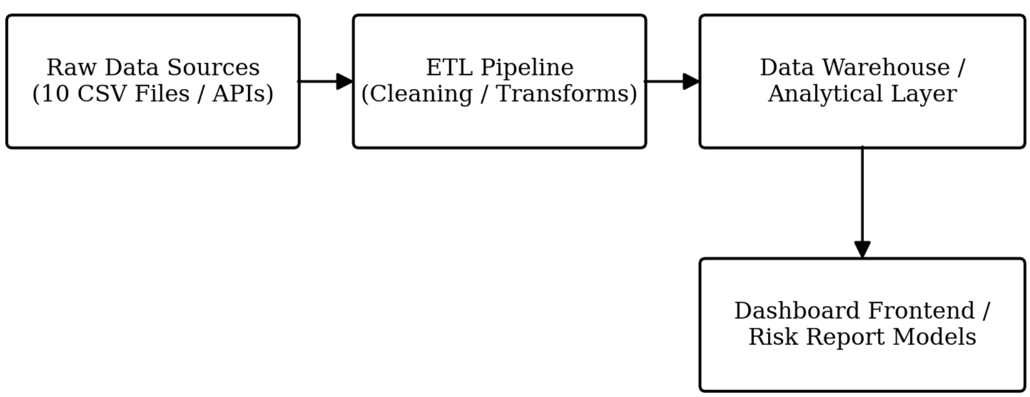
- Schema Validation and Type Enforcement: Parses dates into standard ISO format (YYYY-MM-DD), enforces numeric types on NAVs/AUMs, and strips whitespace from fund identifiers.

- Automated Batch Processing: Scans designated raw data directories, logs ingestion counts, detects schema mismatches, and writes normalized data into the central store.

- Error Handling and Quality Logging: Quarantines corrupt rows (e.g., negative NAV values or missing primary keys) into isolated execution log reports.

Data Engineering and ETL Architecture Design

The ETL system standardizes messy daily data feeds into production-grade analytical tables.



Data Transformations and Calculation Engine

NAV Return Calculations: Computes trailing compound annual growth rates (CAGR) and rolling annualized returns across 1-year, 3-year, and 5-year windows:

$$CAGR = (NAV_{End} / NAV_{Start})^{1/N} - 1$$

- Benchmark Matching: Joins NAV time-series data with benchmark daily returns using standard calendar dates, accounting for market holidays.

- Data Imputation: Fills missing daily values using forward-fill techniques for non-trading days, while dropping inactive funds from active tracking universes.

Quality Assurance Framework

- Duplicate Detection: Identifies duplicate record combinations of (Fund_ID, Date).

- Outlier Scrubbing: Filters out abnormal single-day NAV jumps resulting from stock splits or bad feeds using moving standard deviation bounds.

Exploratory Data Analysis and Risk Metrics

AUM and Cost Efficiency Highlights

- AUM Concentration: Market AUM remains concentrated in top fund houses, though mid-sized AMCs are capturing market share in equity categories.

- Direct vs. Regular Fee Impact: Direct plans consistently outperform Regular counterparts by 0.75% to 1.50% per year, demonstrating the long-term wealth impact of lower expense ratios.

Cumulative Wealth Impact (Hypothetical Rs. 100,000 Investment Over 10 Years)

Scenario	Final Value
Direct Plan (12% Net CAGR)	Rs. 310,585
Regular Plan (10.5% Net CAGR)	Rs. 271,408
Fee Drag Loss	Rs. 39,177

Quantitative Risk Analytics

Tail risk metrics provide deeper insight beyond standard volatility measures:

- Value at Risk (VaR): Measures the maximum expected loss at a 95% confidence interval over a 1-month holding period.
- Conditional Value at Risk (CVaR): Quantifies expected tail losses when market drawdowns breach the VaR threshold.

Asset Category	Avg Std Dev	95% 1-Mo VaR	95% 1-Mo CVaR	Max Drawdown (5-Yr)
Large Cap Equity	14.2%	-6.5%	-9.8%	-22.4%
Mid Cap Equity	18.6%	-8.9%	-13.1%	-28.7%
Small Cap Equity	22.1%	-11.2%	-16.4%	-34.1%
Hybrid Aggressive	10.8%	-4.8%	-7.2%	-15.3%
Short Duration Debt	2.1%	-0.4%	-0.8%	-2.1%

Mutual Fund Performance and Benchmark Analysis

Risk-Adjusted Metrics Engine

To evaluate whether fund managers generate genuine excess returns relative to risk, three core metrics are computed:

Sharpe Ratio evaluates return per unit of total risk:

$$\text{Sharpe} = (R_p - R_f) / \sigma_p$$

Sortino Ratio measures return relative exclusively to downside volatility (σ_d), isolating harmful market movements.

Treynor Ratio measures return generated per unit of systematic market risk (β).

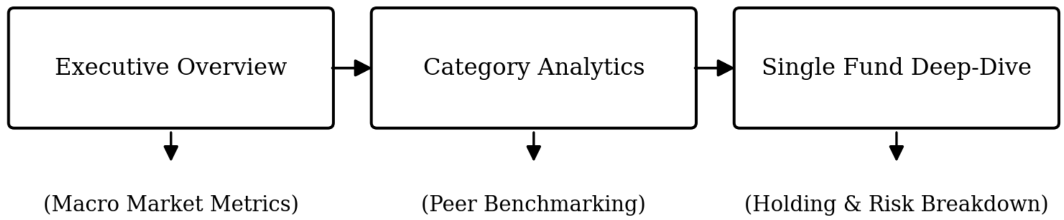
Alpha and Beta Drift

- Benchmark Alpha Generation: Large-cap equity funds demonstrate declining alpha generation over 3-year rolling periods, reinforcing the shift toward low-cost index tracking.
- Style Drift and Beta Instability: Mid-cap and small-cap managers frequently display beta drift relative to standard benchmarks during market drawdowns, shifting allocation toward liquid large-cap holdings.

Interactive Dashboard Architecture and User Flows

Architecture Overview

The system architecture utilizes modular data views designed for seamless navigation:



User Navigation Flows

- Portfolio Managers: Use the peer comparison view to analyze benchmark tracking errors, expense dynamics, and rolling beta shifts.

- Retail Investors and Advisors: Use simple visual cards to compare 3-year risk-adjusted returns (Sortino) against expense ratios between Direct and Regular plans.

Project Limitations and Data Constraints

- Survivorship Bias: Historical datasets exclude merged or liquidated schemes, which can lead to slight overestimates of long-term category average returns.

- Data Frequency Limits: Portfolio holding updates are limited to monthly portfolio disclosures, obscuring intra-month stock turnover and sector shifts.

- System Constraints: Calculating multi-year rolling metrics across daily NAV time-series requires significant memory optimization during batch updates.

Strategic Recommendations and Action Plan

- Strategic Allocation: Shift core large-cap allocations toward passive index funds to eliminate fee drag, reserving active management fees for small-cap and thematic strategies.

- Automated Risk Monitoring: Integrate real-time tracking alerts for funds showing persistent style drift or sudden increases in downside risk (CVaR).

- Pipeline Upgrades: Expand data ingestion script capabilities to support live API integrations for real-time NAV tracking and continuous risk updates.